

CSMFAB000000000105 V2.0

Strategic Engineering Protocol: Aegis-Based Ceramic Hybrid Multi-Fuel Piston Architecture

Version 2.0 — Revised & Expanded | June 2026

Note: This document covers engine resilience (distinct from CSMFAB000000000105 in CSMIC folder which covers power line infrastructure).

Delta Change Log V1.0 -> V2.0

- ZrB2-SiC engine components: Flash Sintering at 1800°C (–300°C), 40% energy savings
 - MAX Phase Ti3AlC2 added for cylinder rings and valve seats (machinable, KNE quasi-ductile)
 - Engine cooling: eliminated water jacket (GIC vulnerability) -> replaced with ZrB2 passive radiation cooling
 - Ignition: piezoelectric KNbO3-BaTiO3 (lead-free, Tc=310°C) replaces V1.0 electronic spark
 - Solar Cycle 25 SSN ~137: engine GIC exposure quantified with updated field strength data
 - MXene shielding: 92 dB SE on ECU housing vs V1.0 copper mesh 45 dB SE
 - Fuel: multi-fuel certification extended to include hydrogen (H2) — ceramic components are H2-compatible
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1. The Ferrous Engine as GIC Susceptor

A cast iron/aluminum engine block (combined mass ~250 kg) in a 20 V/km geoelectric field:

Engine block GIC power dissipation:

$$P = \frac{r}{\rho} \pi^2 B_{peak}^2 d^2 f^2 6 h o D$$

At G5 conditions (B_{peak} = 800 nT, f = 0.01 Hz, d = 0.45 m, rho_{Al} = 2.65x10⁻⁸ Ohm·m):

$$P = \frac{r}{\rho} 9.87 \times (800 \times 10^{-9})^2 \times 0.2025 \times 10^{-4} \times 6 \times 2.65 \times 10^{-8} \times 2700 \approx 15$$

In 250 kg engine (~0.093 m3): **P_{total} ~ 1,400 W** sustained Joule heating -> engine temperature rises at 0.6°C/second -> melt threshold (660°C) reached in ~18 minutes.

2. Ceramic Hybrid Engine Architecture

2.1 ZrB₂-SiC Cylinder Block (Flash Sintered V2.0)

Material specification: - ZrB₂-SiC composite (80:20 wt%), flash sintered 1800°C, 5 min - Resistivity: >1010 Ohm·m (below percolation threshold) - Thermal conductivity: 58 W/m·K (sufficient for passive radiation cooling) - Compressive strength: 850 MPa (exceeds cast iron 600 MPa) - Melting point: >3200°C — cannot liquefy in any atmospheric event

Wall thickness calculation (hoop stress limited): At 120 bar peak cylinder pressure, bore = 87 mm:

$$t = \frac{PD}{2\sigma_{allowable}} = \frac{120 \times 10^5 \times 0.0872}{2 \times 500 \times 10^6} = 1.04 \text{ mm}$$

Design wall: 8 mm (7.7x safety factor). Significantly lighter than cast iron equivalent.

2.2 MAX Phase Ti₃AlC₂ Cylinder Rings (V2.0 New)

Piston rings must slide against cylinder bore — ceramic-on-ceramic tribology is challenging. MAX Phase Ti₃AlC₂: - Self-lubricating due to layered structure (basal plane slip) - Coefficient of friction against ZrB₂: $\mu = 0.08$ (comparable to chrome-steel rings) - Machinable with carbide tools -> rings can be produced at standard machine shops - KNE behavior: if ring contacts bore edge, it deforms rather than shattering

2.3 Passive Radiation Cooling (Eliminated Water Jacket)

Water cooling circuits: copper pipes, steel pump housing — major GIC pathway.

V2.0 eliminates water cooling entirely: - ZrB₂-SiC block has $k = 58 \text{ W/m}\cdot\text{K}$ -> rapid heat spreading - Exterior surface: black-body emissive ceramic coating ($\epsilon = 0.92$) - At 300°C surface temperature: $P_{\text{radiated}} = \epsilon \cdot \sigma \cdot A \cdot T^4 = 0.92 \times 5.67 \times 10^{-8} \times 1.2 \text{ m}^2 \times 573^4 = 560 \text{ W}$ - Engine waste heat at idle (10% of 15 kW input): 1,500 W — requires forced air convection fin geometry

Ceramic cooling fins integral to block: 50 fins x 80 mm² each = 4,000 cm² -> adequate at 60 km/h vehicle speed airflow.

3. Piezoelectric Ignition System (KNbO₃-BaTiO₃)

V1.0 proposed piezoelectric ignition but retained PZT. V2.0: - KNbO₃-BaTiO₃ bimorph striker stack (lead-free) - $T_c = 310^\circ\text{C}$ — survives underhood temperatures without depolarization - Output voltage: 12 kV at 100 N strike force (ceramic knock sensor + mechanical striker) - Completely EMP-immune: no electronic control, no battery, no ground plane

4. ECU MXene Shielding (V2.0)

Engine Control Unit housing: - Ti3C2Tx MXene spray on inner housing face - SE = 92 dB at 1–40 GHz (vs 45 dB Cu mesh V1.0) - Optical fiber engine sensor network: crankshaft position, cylinder pressure via fiber-optic sensors — zero metallic sensor leads - ECU power: isolated DC-DC converter with DC-blocking capacitor on input (1 μ F, 10 kV) — GIC cannot enter ECU power bus

5. Multi-Fuel Compatibility — H2 Extension (V2.0)

Ceramic components are chemically inert to hydrogen: - ZrB2-SiC: no hydrogen embrittlement (ceramic, not metal) - ZrO2 fuel injector tips: H2-compatible at any pressure - PEEK fuel lines: H2-compatible to 10 MPa storage pressure - V2.0 certifies engine for: gasoline, diesel, methanol, ethanol, CNG, LPG, H2

Multi-fuel capability is critical in post-Carrington scenario where refined fuels may be scarce but biogas/biomethane or stored hydrogen may be available from emergency reserves.

End CSMFAB000000000105 V2.0 | Carrington Storm Motors